

ANMAYA AGARWAL

+1 (347) 216-2308

<https://www.linkedin.com/in/anmaya-agarwal/>

anagarwa@andrew.cmu.edu

EDUCATION

CARNEGIE MELLON UNIVERSITY, TEPPER SCHOOL OF BUSINESS

Pittsburgh, PA

Master of Science in Computational Finance – MSCF

CGPA: 3.63/4.00

12/26

- Received the MSCF Distinguished Merit Scholarship.
- Coursework:** Fixed Income, Financial Data Science, Stochastic Calculus, Machine Learning and Options.

NMIMS UNIVERSITY

Mumbai, India

Master of Business Administration in Technology Management – MBA

CGPA: 3.85/4.00

05/24

- Achieved Dean's List recognition (**top 10% of the class**) for meritorious performance and academic excellence.
- Skills:** Python (Intermediate), Data Visualization: PowerBI (Intermediate), SQL (Basic), Excel & PowerPoint.

Bachelor of Technology in Computer Engineering

- Leadership:** Co-Founded the CodeChef Society for Competitive Programming which increased to 100+ students.

EXPERIENCE

INDIA GOLD POLICY CENTRE, IIM AHMEDABAD

Ahmedabad, India

Research Assistant

07/24 – 04/25

- International Gold Trade Analysis:** Co-authored an IIMA working paper titled “*Strategic Acquisition and Value Addition of Gold Resources for India*,” which proposes strategic solutions to **reduce India's Current Account Deficit** for gold by analyzing granular **8-digit trade classification data** on gold and gold-extraction-related items from Japan and Switzerland.
- Risk Management Case Study:** Co-authored an IIMA case study “*The Rise and Fall of Bullion Banking at Scotiabank*,” examining management, reputation and derivatives exposure failures in precious metals trading and banking.
- Data Infrastructure Development:** Designed a comprehensive Gold Data Hub by automating the extraction of data from 8+ sources using Python, enhancing visualization through Power BI dashboards, and **saving 3–5 hours of manual work** weekly.
- Achievements:** Earned 2024-25 IIMA Endowment Fund Case Award and Best Research Presentation Award at IMRC 2024.

PIRAMAL CAPITAL AND HOUSING FINANCE LTD.

Mumbai, India

Data Analyst Intern

07/23 – 09/23

- Multifactor Sectoral Forecasting:** Developed regression-based factor models for **7 sectoral indices**, leveraging Granger Causality to identify leading macroeconomic indicators and isolate significant predictive signals from noise.
- Dimensionality Reduction:** Implemented a systematic feature selection pipeline using Correlation Analysis and Multicollinearity thresholds (VIF), **reducing the feature space by approximately 80-90%** to mitigate overfitting.

COMPETITIONS & PROJECTS

BRIDGEWATER X METACULUS FORECASTING CONTEST 2026 (*Overall Rank 55/2867*)

03/26

- Probabilistic Forecasting:** Ranked 55th globally (**Top 2 Percentile**) by modeling accurate probability distributions for macroeconomic events; continuously calibrated predictions to account for incoming market data and geopolitical signals.

ROTMAN INTERNATIONAL TRADING COMPETITIONS

02/26 & 09/25

- Algorithmic Market Making:** **Ranked 10th globally** by engineering a high-frequency trading strategy using asyncio to minimize API latency; implemented dynamic bid-ask skewing based on micro-price signals and inventory risk while managing exposure through a real-time shadow order tracking system.
- Algorithmic ETF Arbitrage:** Achieved **Rank 1/66 at CMU** by combining **ETF arbitrage with market making**, executing dynamic price quoting based on fair value, inventory risk, and market microstructure signals like order book imbalance.
- Volatility Trading:** Identified mispricing opportunities between analyst forecasts and market implied volatility of options to **profit from volatility differentials** while managing portfolio risk within delta limits.

ALGORITHMIC DEBT SIMPLIFICATION ENGINE (*GitHub Link*)

12/25

- Performance Benchmarking:** Engineered an approximation for a variation of the NP-Complete “Zero-Sum Set Packing” problem, **benchmarked performance across four solvers** including Mixed Integer Linear Programming (Gurobi), Randomized Greedy (Monte Carlo), Simple Greedy, and a custom Layered Approximation heuristic.
- Optimization & App Integration:** Achieved sub-second latency on 10,000+ simulated users with >5% superior edge minimization over greedy baselines, integrating the solver into the backend of “Divy” (live iOS/Android app).

ADDITIONAL INFORMATION

- Interests:** Solving Puzzles (Jane Street Feb 2026); Squash (National level); Landscape and Wildlife Photography.
- Certificates:** Bloomberg Market Concepts; Akuna Capital Options 201; Akuna Capital Options 101; National Institute of Securities Markets (NISM) Series XV Research Analyst & Series VIII Equity Derivatives.